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Department of Computer Science and Engineering.
Academic Year: 2023 - 2024(Odd/~~Even~~ Semester)

Degree, Semester & Branch: B.E, III & EEE
Course Code & Title: CS3353 & C programming and Data Structure
Name of the Faculty member (s): A. Sofia Thanga Mary, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit III / Application of stack**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO8
  - **Activity Chosen:** Think pair and share
  - **Justification:** After explaining the concept of evaluation of expression such as infix to postfix conversion and postfix evaluation in stack application, to understand the learning level of the student about evaluating the expression this activity was given.
  - **Time Allotted for the Activity:** 15 Minutes
  - **Details of the Implementation:**
    - Displayed the different questions in the topics infix to postfix conversion and postfix evaluation for each pair of students in the class as shown in Figure 1.
    - Identified 32 pair of students and asked them to think for five minute about evaluation of expression.
    - After that, the students were asked to discuss the answer for the questions in the Dias as shown in the figure 2,3.
  - **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1	PSO2
	3	3	2	2	1	1	2	1	2	2

- **PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PSO1**	**PSO2**
	3	**3**	**2**	**2**	**2**	**2**
Justification for correlation	Students will be able to apply linear data structure concepts to identify solution to complex engineering problems	Able to Identify, and analyse engineering problems with queue and stack data structures concept	Able to design solutions for complex engineering problems using different data structures operations	Able to provide complex problem solutions with relevant data structure concepts and operations	Students can be able to apply linear data structures concept to solve various application in modern software packages	Students will be able to specify the electronic system with linear data structure

• Images / Screenshot of the practice:

Applications of Stack

FIG 1: Questions given to students

Fig 2: Discuss with their pair **Fig 3: A student share answer**

Reflective Critique:

- **Feedback of practice from students and other stakeholders:**
  - The students are actively participated in the activity.
  - The student said this activity helped to think and recall information and improved our problem solving skills.

- **Benefit of the practice:**
  - The majority of students correctly answered the question that was given to them.
  - When students communicated to their peers, their nervousness was reduced.
  - It helps the students to think individually about a topic or answer to a given question.
  - With the help of this activity, each student can review the idea of evaluation of expression and infix to postfix conversion and gain a thorough knowledge of the topic.

- **Challenges faced in implementation**:

  Most of the students are hesitant to discuss the ideas they have learned in the classroom.

- **References**:

1. https://www.readingrockets.org/strategies/think-pair-share
2. [https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-canimprove-your-classroom-discussions 1704 .html](https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-canimprove-your-classroom-discussions1704.html)

Signature of Faculty Member **HOD**